

OASIS INFOBYTE
DATA ANALYTICS INTERNSHIP
LEVEL 1 – TASK 1
EXPLORATORY DATA ANALYSIS ON RETAIL SALES DATA

Final Project Report

Submitted By

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Internship

OASIS INFOBYTE – Data Analytics Internship

Domain

Data Analytics

Project Title

Level 1 – Task 1: Exploratory Data Analysis on Retail Sales Data

Technologies Used

Python • Pandas • NumPy • Matplotlib • Seaborn • Jupyter Notebook

Modules Implemented

Dataset Loading • Data Inspection • Descriptive Statistics • Monthly & Quarterly Sales Analysis
Customer Age & Gender Analysis • Top 10 Product Analysis • Category Revenue Analysis
Correlation Analysis • Profit Margin Analysis • Data Visualization

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Year

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to the internship program for providing me with the opportunity to work on the Level 1 – Task 1 project, Exploratory Data Analysis on Retail Sales Data. This project provided practical exposure to data loading, data inspection, descriptive statistics, time-series analysis, customer analysis, product analysis, correlation analysis, visualization, and business interpretation using Python.

I would also like to thank the mentors and coordinators for providing the project requirements and learning resources. The structured task helped me understand how exploratory data analysis can be used to convert raw retail transaction data into meaningful business insights.

During this project, I worked with a retail sales dataset containing 113,036 records and 18 variables. The analysis covered monthly and quarterly revenue trends, customer age-group and gender distribution, top-selling products, product-category revenue, correlation among numerical variables, and category-level profit margins.

Finally, I thank everyone who directly or indirectly supported the successful completion of this project. The experience strengthened my practical understanding of Python-based data analytics and business-oriented interpretation of data.

ABSTRACT

This project focuses on Exploratory Data Analysis (EDA) of a retail sales dataset using Python. The main objective is to inspect the dataset, understand its structure and numerical variables, identify sales and customer patterns, and derive actionable business insights through statistical analysis and visualization.

The dataset contains 113,036 records and 18 variables, including transaction date, customer age, age group, gender, country, state, product category, product, order quantity, unit cost, unit price, profit, cost, and revenue. Pandas and NumPy were used for data handling and statistical calculations, while Matplotlib and Seaborn were used to create visualizations.

The analysis includes monthly and quarterly revenue trends, customer distribution by age group and gender, top 10 best-selling products, revenue by product category, correlation analysis, and profit-margin comparison across product categories. The results show strong revenue contribution from Bikes, high profit margins for Accessories, strong demand for selected products, and strong relationships among financial variables such as Cost, Revenue, and Profit.

The project demonstrates how EDA can support inventory planning, marketing decisions, product prioritization, and profitability improvement. The findings provide a clear foundation for further customer segmentation, predictive analytics, and dashboard development.

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CHAPTER 1: INTRODUCTION

1.1 Project Overview

The Exploratory Data Analysis on Retail Sales Data project focuses on understanding a large retail transaction dataset and extracting meaningful patterns from it using Python. The analysis begins with data loading and inspection and progresses through descriptive statistics, time-series analysis, customer analysis, product analysis, correlation analysis, and profitability analysis.

The project uses the Customer Segmentation / retail sales CSV dataset available in the project folder. The notebook implementation uses Pandas for data manipulation, NumPy for numerical operations, Matplotlib for plotting, and Seaborn for the correlation heatmap. The analysis is performed in a Jupyter Notebook environment.

1.2 Problem Statement

Raw retail transaction data can contain a large number of records and variables, making it difficult to identify important business patterns directly. The problem addressed in this project is to transform the transaction-level data into understandable summaries and visualizations that reveal sales trends, customer composition, product demand, category performance, and relationships between financial variables.

1.3 Project Objectives

1. Load and inspect the retail sales dataset.
2. Understand dataset dimensions, column names, data types, and missing values.
3. Calculate descriptive statistics for numerical variables.
4. Perform monthly and quarterly sales trend analysis.
5. Analyze customer distribution by age group and gender.
6. Identify the top 10 best-selling products.
7. Compare revenue and profitability across product categories.
8. Generate a correlation heatmap for numerical variables.
9. Create an additional visualization showing category-level profit margins.
10. Interpret the results and provide actionable business recommendations.

CHAPTER 2: DATASET & TOOLS

2.1 Dataset Description

The project dataset contains 113,036 transaction records and 18 variables. The columns used in the analysis include Date, Day, Month, Year, Customer_Age, Age_Group, Customer_Gender, Country, State, Product_Category, Sub_Category, Product, Order_Quantity, Unit_Cost, Unit_Price, Profit, Cost, and Revenue. These fields support analysis of customer characteristics, product demand, revenue, cost, and profitability.

2.2 Tools & Technologies Used

Tool / Technology	Purpose
Python	Programming and data analysis
Pandas	Data loading, transformation, grouping and aggregation
NumPy	Numerical operations
Matplotlib	Sales and business visualizations
Seaborn	Correlation heatmap visualization
Jupyter Notebook	Interactive implementation and documentation
CSV Dataset	Input retail transaction data

2.3 Project Folder Structure

The project was organized into separate folders for the dataset, notebook implementation, output files, report documentation, and screenshots. The structure supports easy navigation and keeps the implementation and evidence files organized.

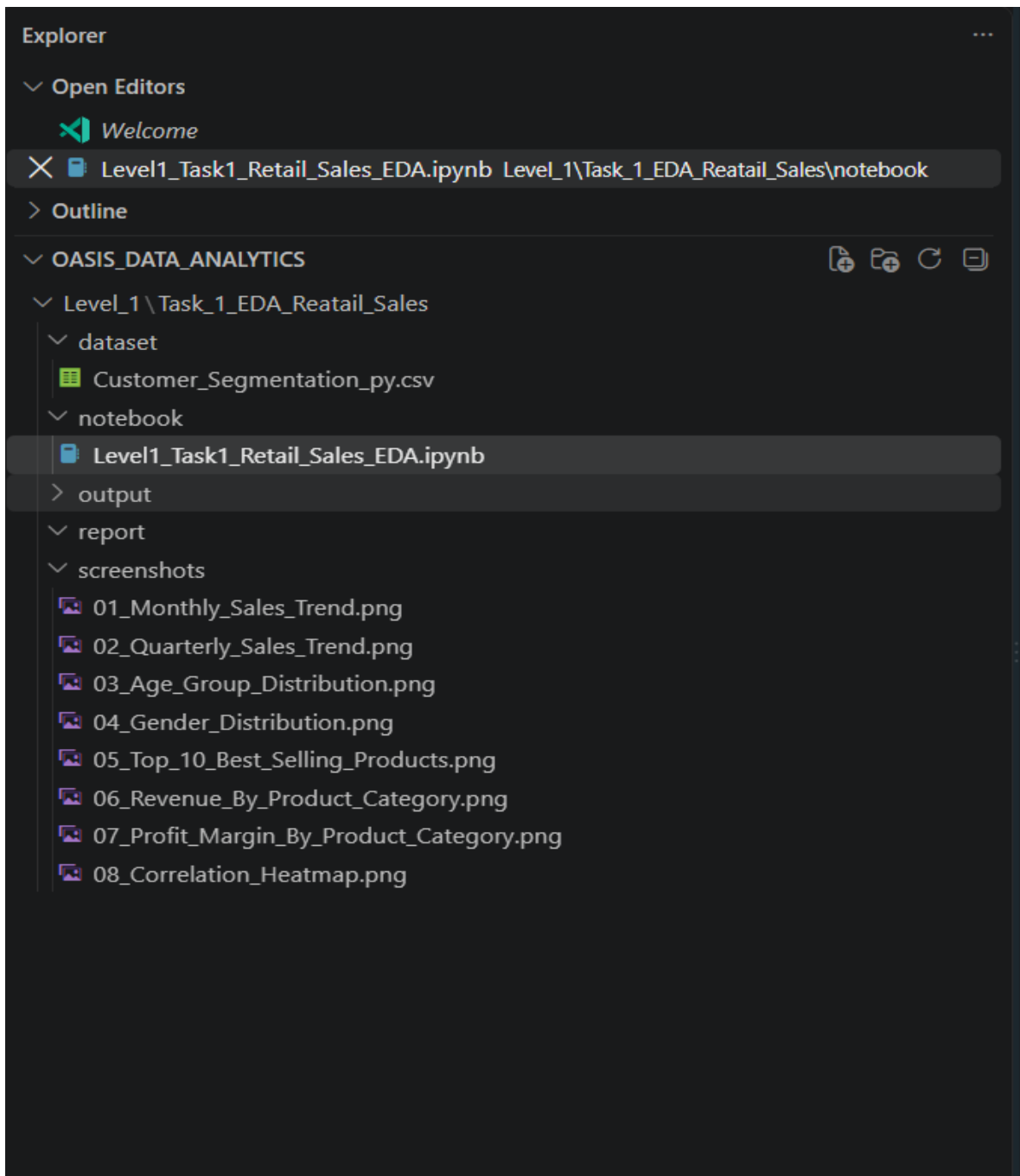


Figure 2.1: Project folder structure used for the Level 1 – Task 1 implementation.

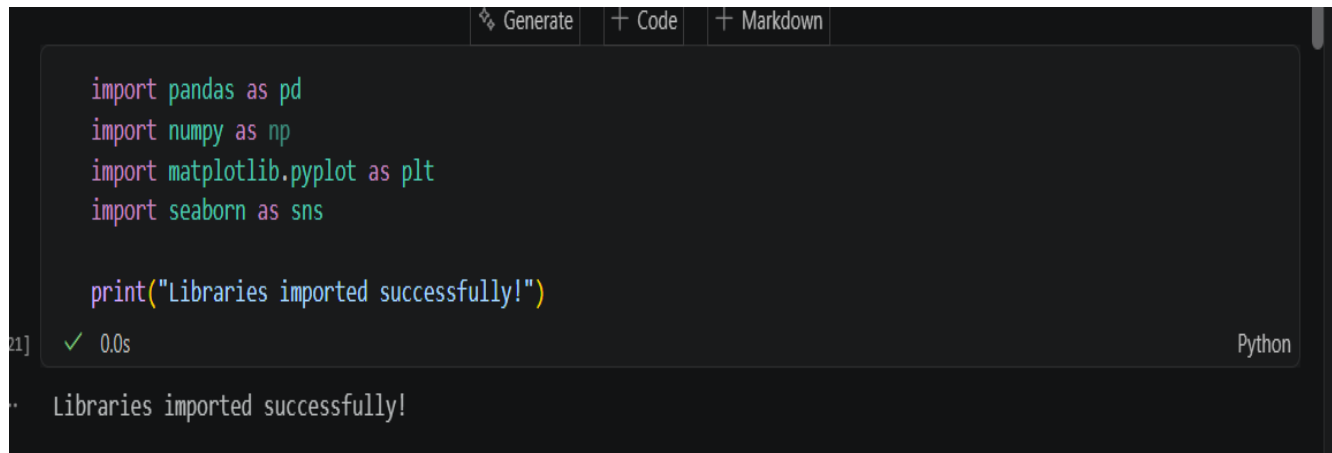
2.4 Input Dataset

The input file used in the notebook is `Customer_Segmentation_py.csv`. It is loaded with Pandas using the relative path `../dataset/Customer_Segmentation_py.csv`. The dataset contains transaction-level retail information and is used for all subsequent EDA steps.

CHAPTER 3: EDA IMPLEMENTATION

3.1 Library Import & Dataset Loading

The required Python libraries were imported successfully. Pandas was used for DataFrame operations, NumPy for numerical support, Matplotlib for charts, and Seaborn for the correlation heatmap. The CSV dataset was then loaded into a Pandas DataFrame.



```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries imported successfully!")
```

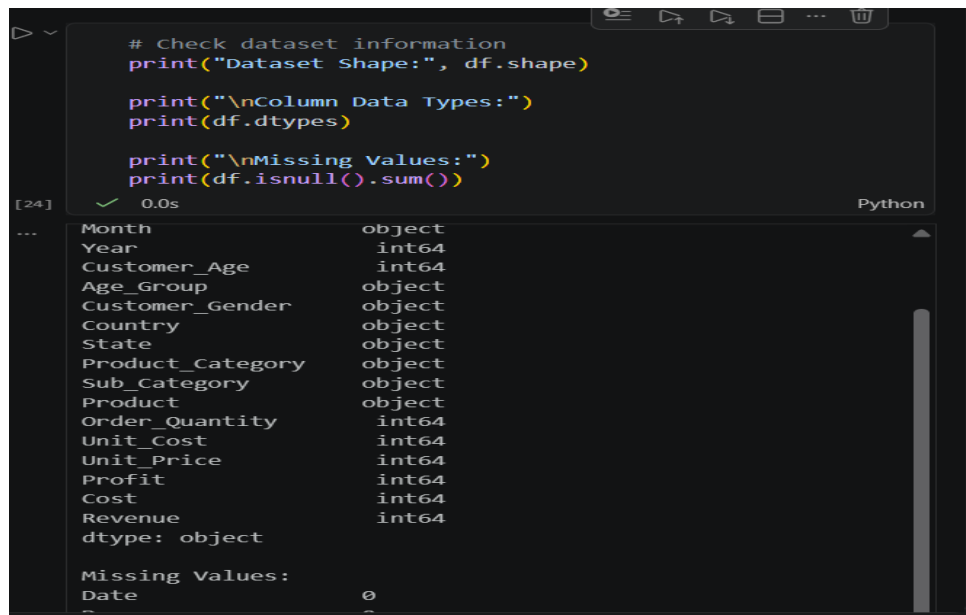
21] ✓ 0.0s Python

Libraries imported successfully!

Figure 3.1: Python Library Import and Dataset Loading

3.2 Dataset Inspection

The dataset was inspected using shape, data types, missing-value counts, and df.info(). This step confirmed the size and structure of the data before analysis.



```
# Check dataset information
print("Dataset Shape:", df.shape)

print("\nColumn Data Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())
```

[24] ✓ 0.0s Python

Month	object
Year	int64
Customer_Age	int64
Age_Group	object
Customer_Gender	object
Country	object
State	object
Product_Category	object
Sub_Category	object
Product	object
Order_Quantity	int64
Unit_Cost	int64
Unit_Price	int64
Profit	int64
Cost	int64
Revenue	int64
dtype:	object
Missing Values:	
Date	0
~	~

Figure 3.2: Dataset Inspection Showing Data Types and Missing Values

Interpretation:

The inspection confirmed that the dataset contains 18 variables with numerical and categorical fields suitable for EDA. The analysis could therefore proceed to statistical and visual exploration.

3.3 Descriptive Statistics

Descriptive statistics were calculated for all numerical variables. Mean, median, mode, and standard deviation were used to understand central tendency and variability.

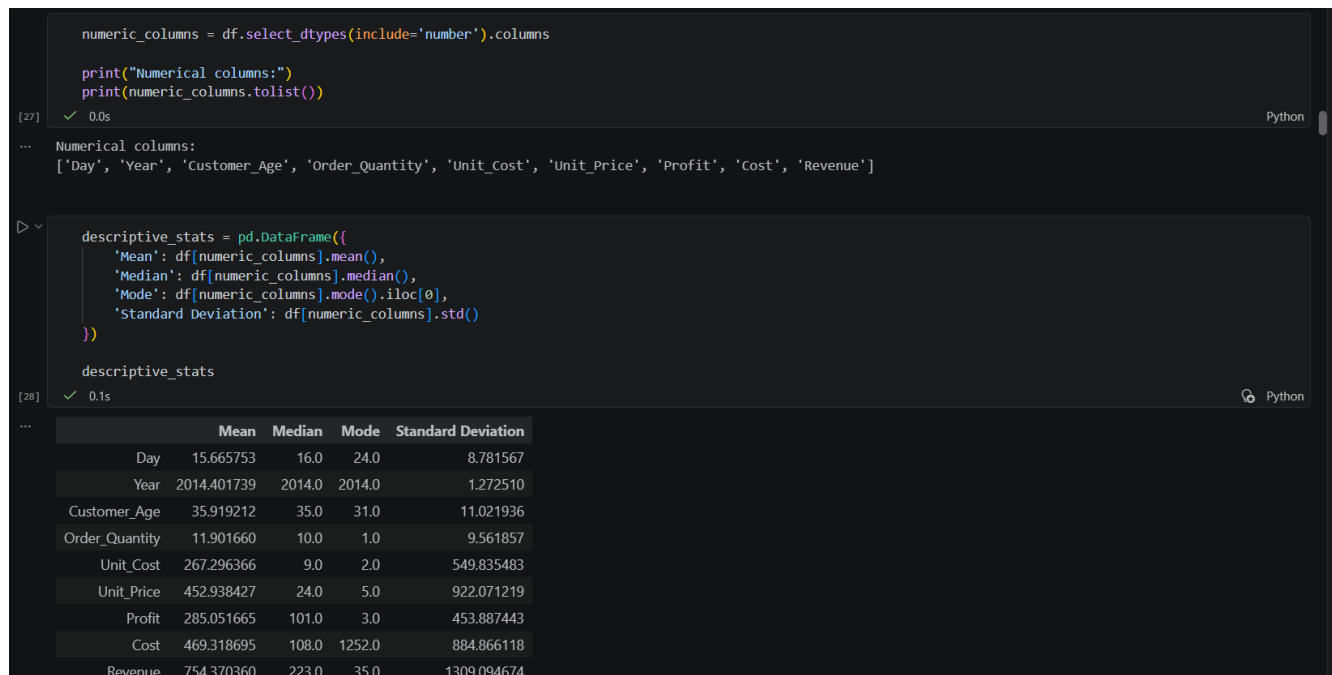


Figure 3.3: Descriptive Statistics of Numerical Variables

Interpretation:

The resulting table summarizes the distribution of Day, Year, Customer_Age, Order_Quantity, Unit_Cost, Unit_Price, Profit, Cost, and Revenue. Differences between means and medians indicate variation and skewness in several business variables.

3.4 Monthly Sales Trend

The Date column was converted to datetime format and grouped by month. Revenue was then aggregated for each month to identify seasonal fluctuations and changes in sales performance.

```

df['Year_Month'] = df['Date'].dt.to_period('M')

monthly_revenue = df.groupby('Year_Month')['Revenue'].sum()

monthly_revenue.head()

```

[30] ✓ 0.0s

```

... Year_Month
2011-01    675193
2011-02    637598
2011-03    708517
2011-04    698782
2011-05    734537
Freq: M, Name: Revenue, dtype: int64

```

```

plt.figure(figsize=(14, 6))

monthly_revenue.plot(kind='line', marker='o')

plt.title('Monthly Sales Trend')
plt.xlabel('Month')
plt.ylabel('Total Revenue')
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()

plt.show()

```

[31] ✓ 0.3s

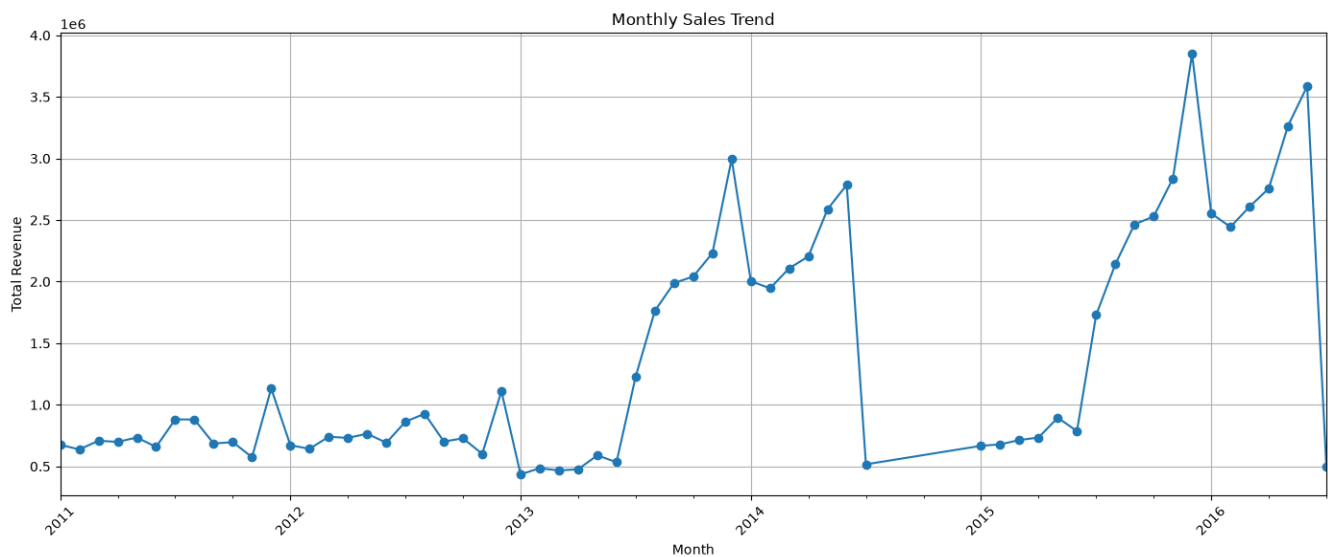


Figure 3.4: Monthly Sales Trend of Retail Revenue

Interpretation:

Revenue fluctuates across the years, with stronger peaks in later periods. The pattern indicates seasonal variation and overall improvement in sales performance, which can support inventory and marketing planning.

3.5 Quarterly Sales Trend

Quarterly revenue was calculated by converting the Date column into quarterly periods and aggregating Revenue for each quarter.

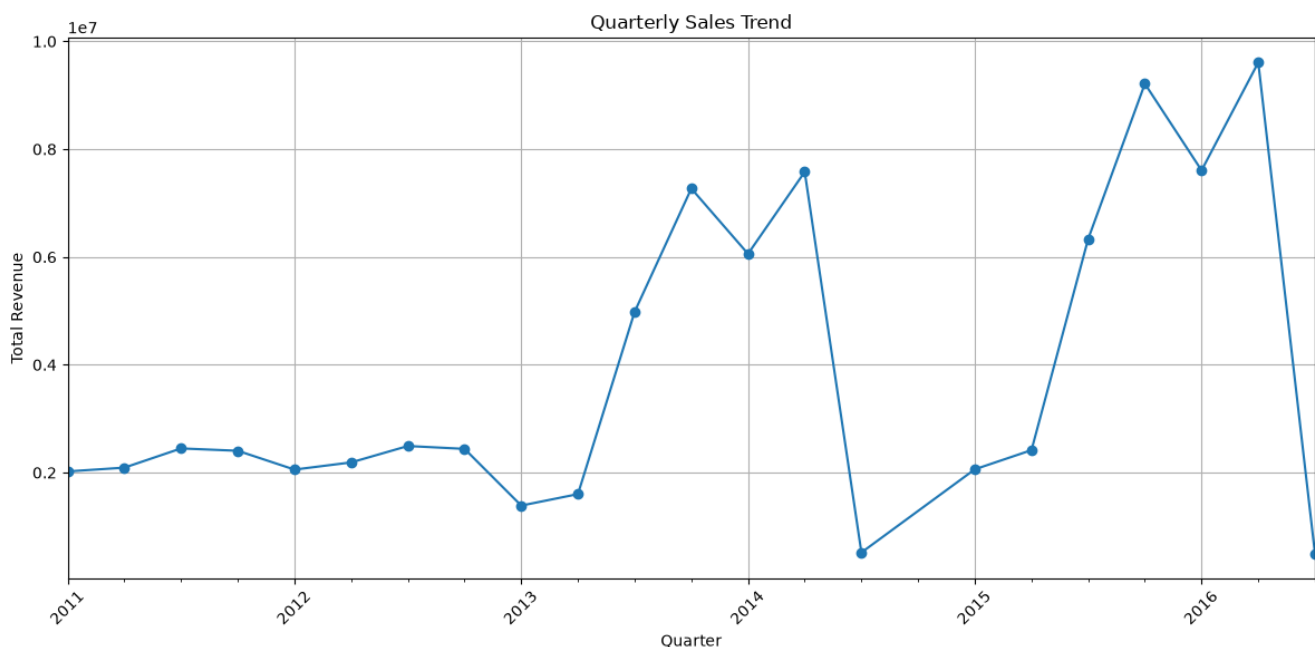
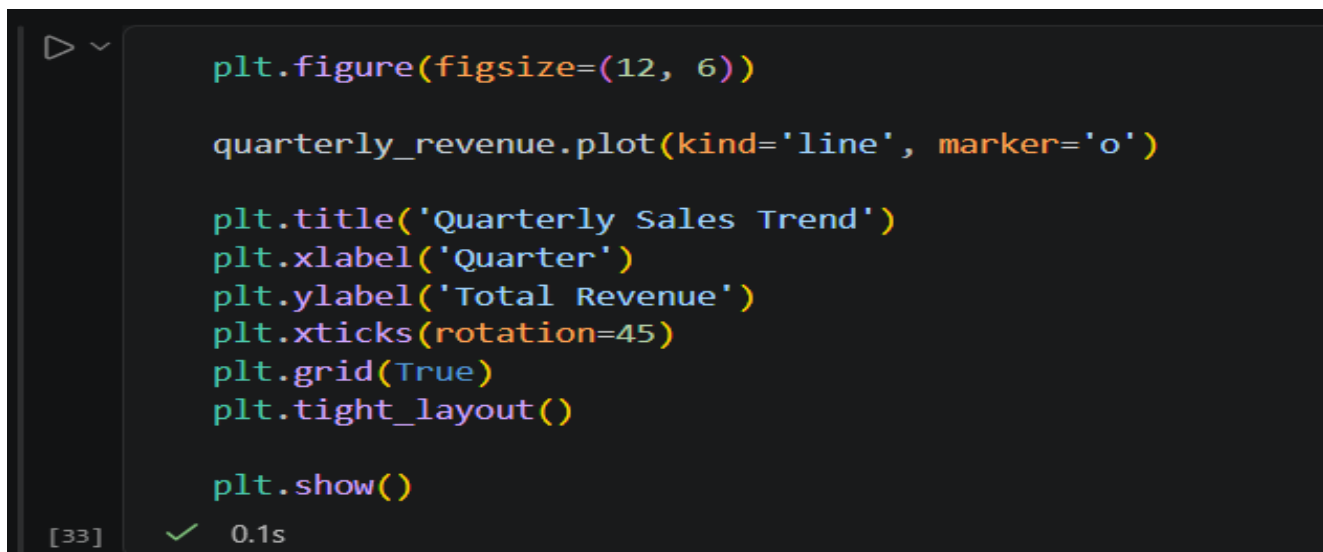


Figure 3.5: Quarterly Sales Trend

Interpretation:

Quarterly revenue varies considerably, with much stronger performance in later periods. The recurring rises and declines suggest that demand is not uniform across quarters and should be considered when planning stock and campaigns.

3.6 Customer Analysis

Customer analysis was performed using age-group and gender distributions. These analyses help identify the dominant customer segments and provide a basis for targeted marketing decisions.

3.6.1 Age Group Distribution

```
[34] ✓ 0.0s

print("Age Groups:")
print(df['Age_Group'].value_counts())

... Age Groups:
Age_Group
Adults (35-64)      55824
Young Adults (25-34) 38654
Youth (<25)         17828
Seniors (64+)        730
Name: count, dtype: int64

age_group_counts = df['Age_Group'].value_counts()

plt.figure(figsize=(10, 6))

age_group_counts.plot(kind='bar')

plt.title('Customer Distribution by Age Group')
plt.xlabel('Age Group')
plt.ylabel('Number of Customers')
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.tight_layout()

plt.show()

[35] ✓ 0.1s
```

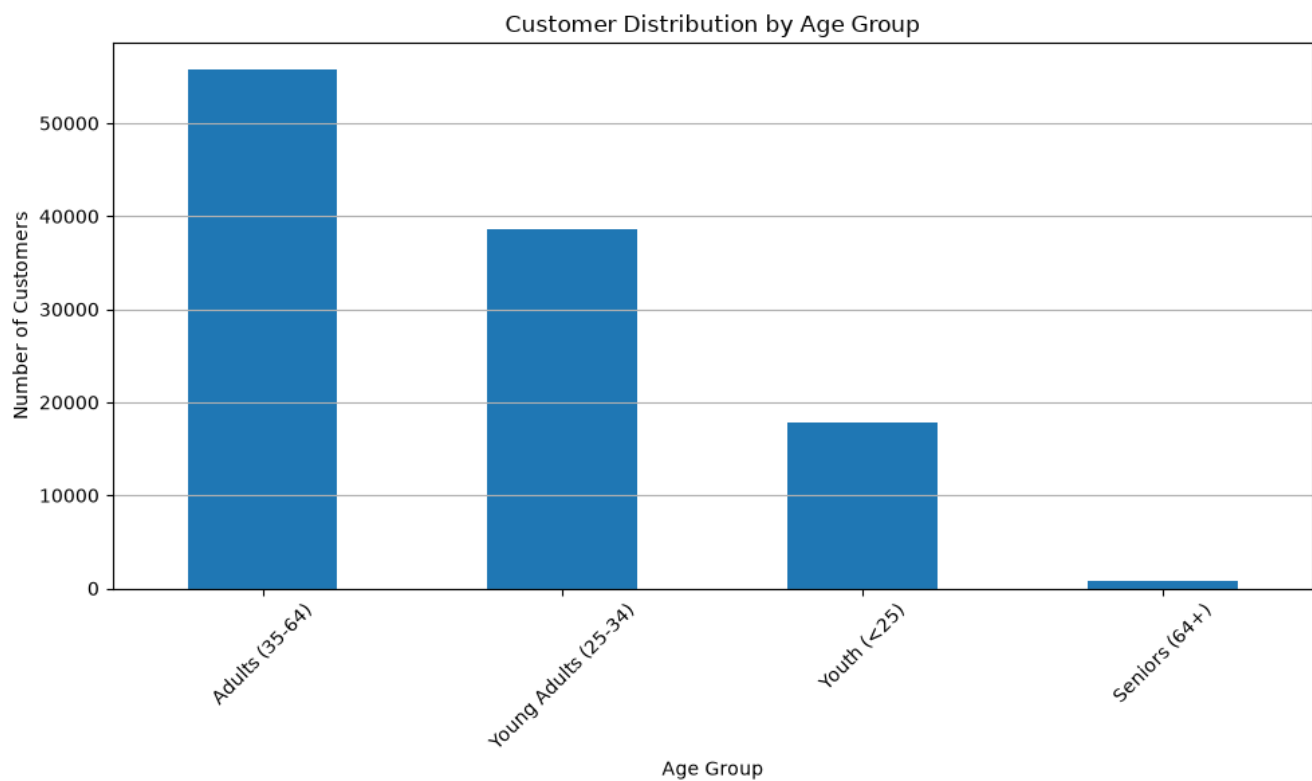


Figure 3.6.1: Customer Distribution by Age Group

Interpretation:

Adults (35–64) form the largest segment with 55,824 records, followed by Young Adults (25–34) with 38,654. Youth account for 17,828 records, while Seniors (64+) are the smallest segment with 730 records.

3.6.2 Gender Distribution

```

gender_counts = df['Customer_Gender'].value_counts()

print("Customer Gender Distribution:")
print(gender_counts)

[36] ✓ 0.0s

... Customer Gender Distribution:
Customer_Gender
M      58312
F      54724
Name: count, dtype: int64

plt.figure(figsize=(8, 6))

gender_counts.plot(kind='bar')

plt.title('Customer Distribution by Gender')
plt.xlabel('Gender')
plt.ylabel('Number of Customers')
plt.xticks(rotation=0)
plt.grid(axis='y')
plt.tight_layout()

plt.show()

[37] ✓ 0.1s

```

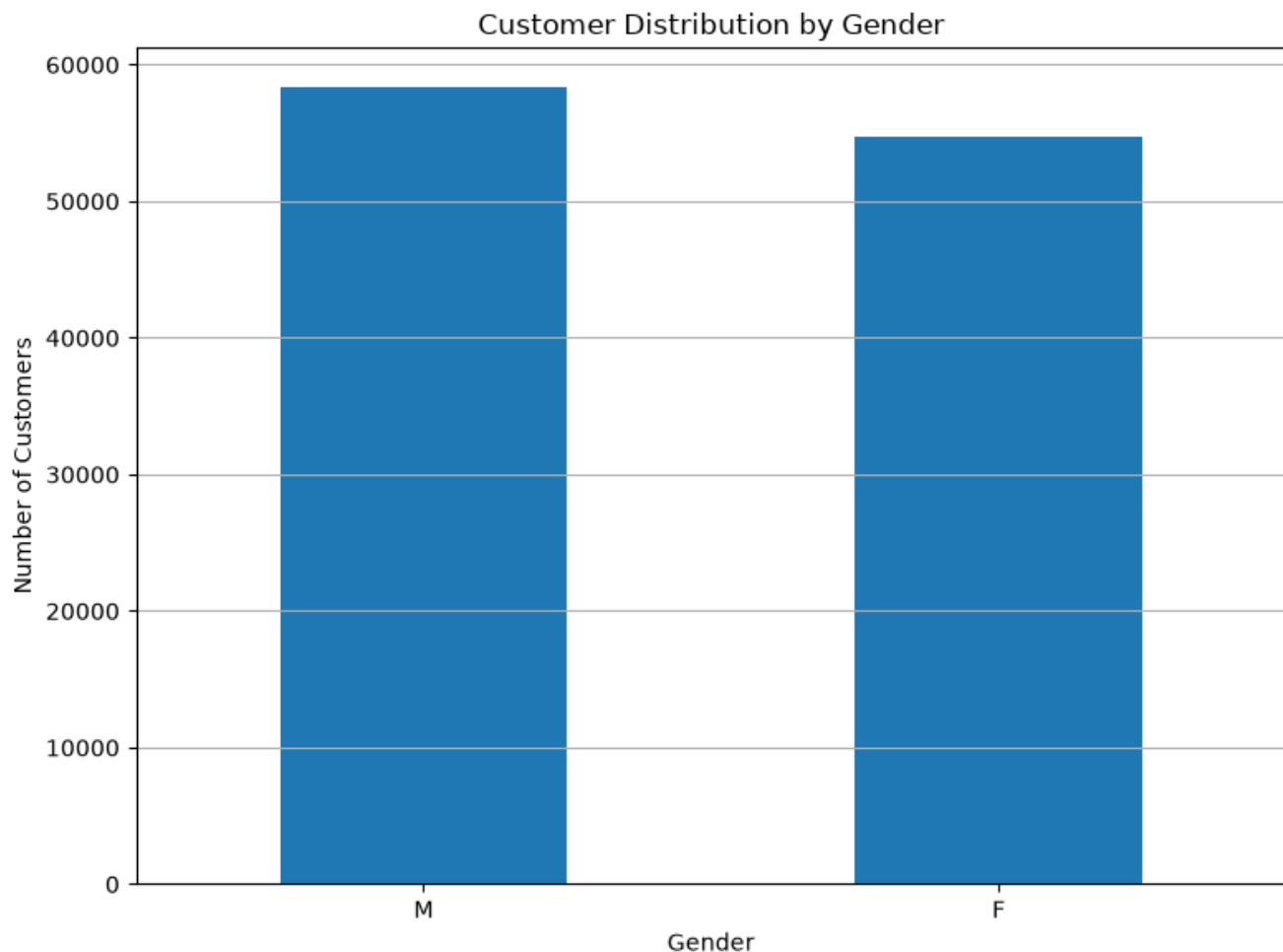


Figure 3.6.2: Customer Distribution by Gender

Interpretation:

The customer base is relatively balanced by gender. Male customers account for 58,312 records and female customers account for 54,724, indicating that both groups are important target audiences.

3.7 Product Analysis

3.7.1 Top 10 Best-Selling Products

Products were grouped by Product and ranked using the total Order_Quantity. The top ten products were selected to identify high-demand items.

```
    .head(10)
)

print("Top 10 Best-Selling Products:")
print(top_products)
```

[39] ✓ 0.0s

```
... Top 10 Best-Selling Products:
Product
Water Bottle - 30 oz.      164086
Patch Kit/8 Patches       157583
Mountain Tire Tube        102792
AWC Logo Cap              67316
Sport-100 Helmet, Red     63663
Road Tire Tube            62296
Fender Set - Mountain     62118
Sport-100 Helmet, Black   62105
Touring Tire Tube         56802
Sport-100 Helmet, Blue    55895
Name: Order_Quantity, dtype: int64
```

```
plt.figure(figsize=(12, 6))

top_products.sort_values().plot(kind='barh')

plt.title('Top 10 Best-Selling Products')
plt.xlabel('Total Quantity Sold')
plt.ylabel('Product')
plt.grid(axis='x')
plt.tight_layout()

plt.show()
```

[40] ✓ 0.2s

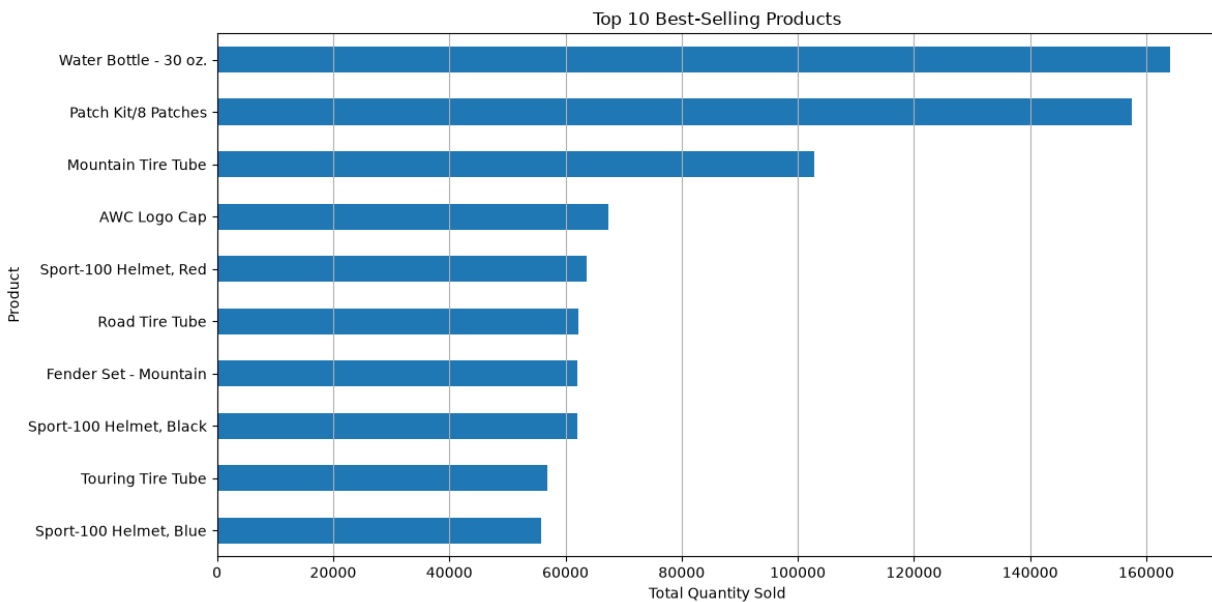


Figure 3.7.1: Top 10 Best-Selling Products by Quantity

Interpretation:

Water Bottle – 30 oz. is the top-selling product with 164,086 units, followed by Patch Kit/8 Patches with 157,583 units. Mountain Tire Tube ranks third with 102,792 units. These high-demand products should receive consistent inventory support.

3.7.2 Revenue by Product Category

```
category_revenue = (
    df.groupby('Product_Category')['Revenue']
      .sum()
      .sort_values(ascending=False)
)

print("Revenue by Product Category:")
print(category_revenue)
```

```
[41] ✓ 0.0s
```

```
... Revenue by Product Category:
Product_Category
Bikes             61782134
Accessories       15117992
Clothing           8370882
Name: Revenue, dtype: int64
```

```
plt.figure(figsize=(10, 6))

category_revenue.plot(kind='bar')

plt.title('Revenue by Product Category')
plt.xlabel('Product Category')
plt.ylabel('Total Revenue')
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.tight_layout()

plt.show()
```

```
[42] ✓ 0.1s
```

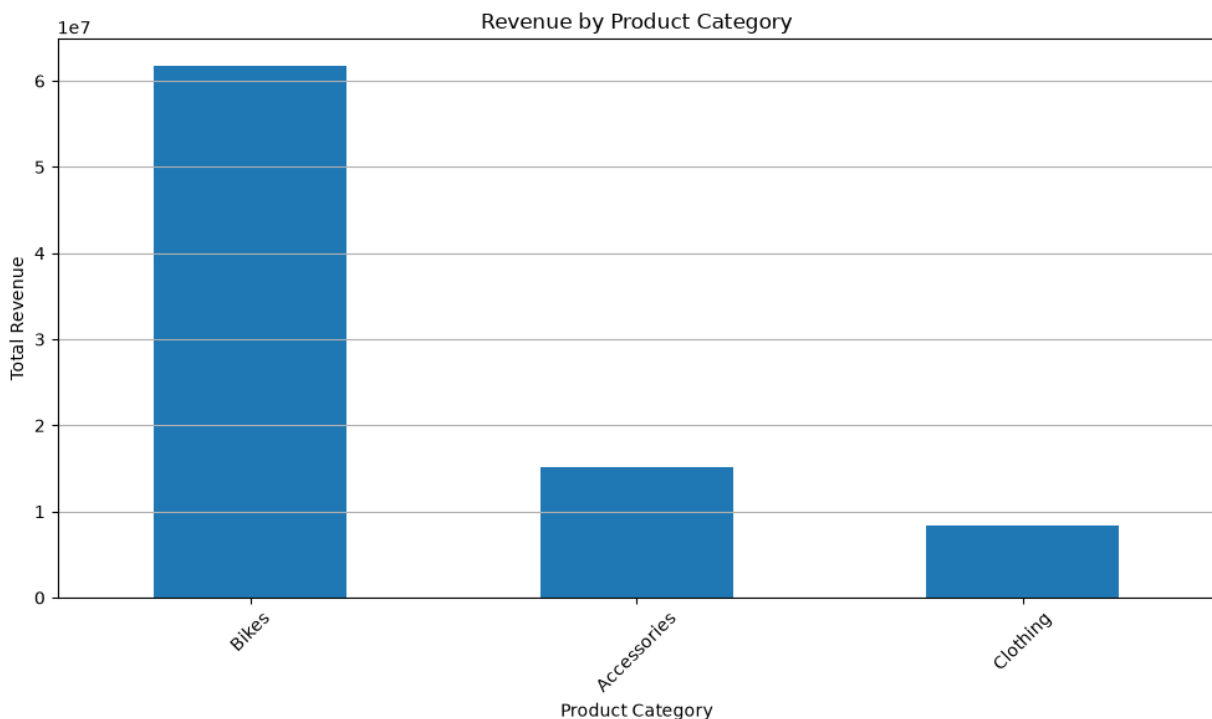


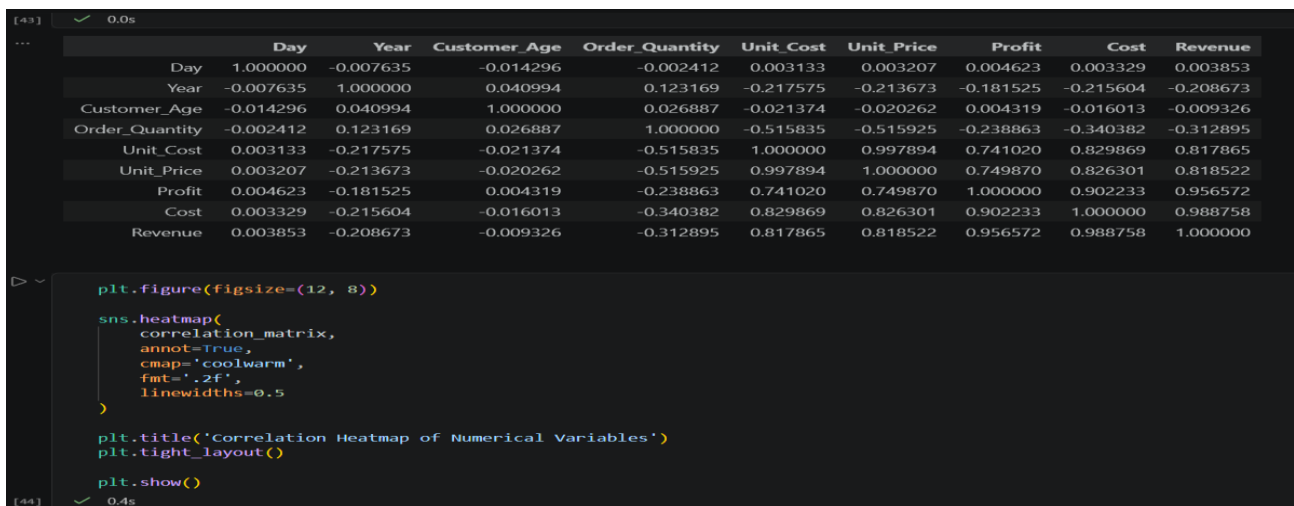
Figure 3.7.2: Revenue by Product Category

Interpretation:

Bikes generate the highest revenue at 61,782,134, followed by Accessories at 15,117,992 and Clothing at 8,370,882. Bikes are therefore the primary revenue-generating category.

3.8 Correlation Analysis

A correlation matrix was calculated for the numerical variables and visualized as a heatmap. This identifies strong positive or negative linear relationships between financial and operational variables.



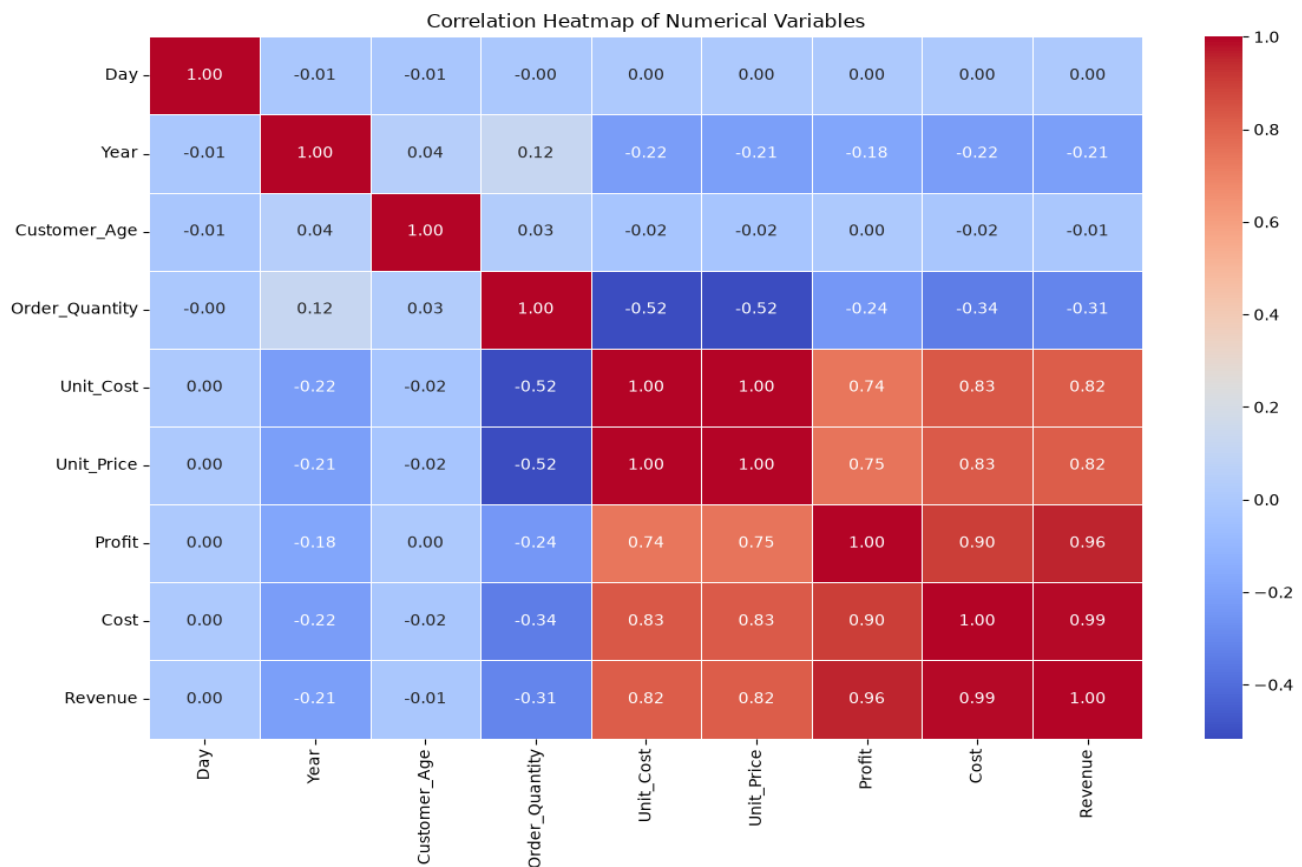


Figure 3.8: Correlation Heatmap of Numerical Variables

Interpretation:

Cost and Revenue have the strongest positive correlation at 0.99, followed by Profit and Revenue at 0.96 and Profit and Cost at 0.90. Unit Cost and Unit Price are also strongly related to Revenue, while Order Quantity has moderate negative relationships with Unit Cost and Unit Price.

3.9 Additional Visualization – Profit Margin by Product Category

As an additional visualization, category-level profit margins were calculated from total Profit divided by total Revenue. This reveals an important business insight that cannot be obtained from revenue totals alone.

```

category_summary['Profit_Margin_%'] = (
    category_summary['Profit'] / category_summary['Revenue']
) * 100

category_summary = category_summary.sort_values(
    'Profit_Margin_%',
    ascending=False
)

category_summary
[47] ✓ 0.0s

...

```

Product_Category	Revenue	Profit	Profit_Margin_%
Accessories	15117992	8862377	58.621390
Clothing	8370882	2839447	33.920524
Bikes	61782134	20519276	33.212313

```

plt.figure(figsize=(10, 6))

category_summary['Profit_Margin_%'].plot(kind='bar')

plt.title('Profit Margin by Product Category')
plt.xlabel('Product Category')
plt.ylabel('Profit Margin (%)')
plt.xticks(rotation=45)
plt.grid(axis='y')
plt.tight_layout()

plt.show()
[48] ✓ 0.1s

```

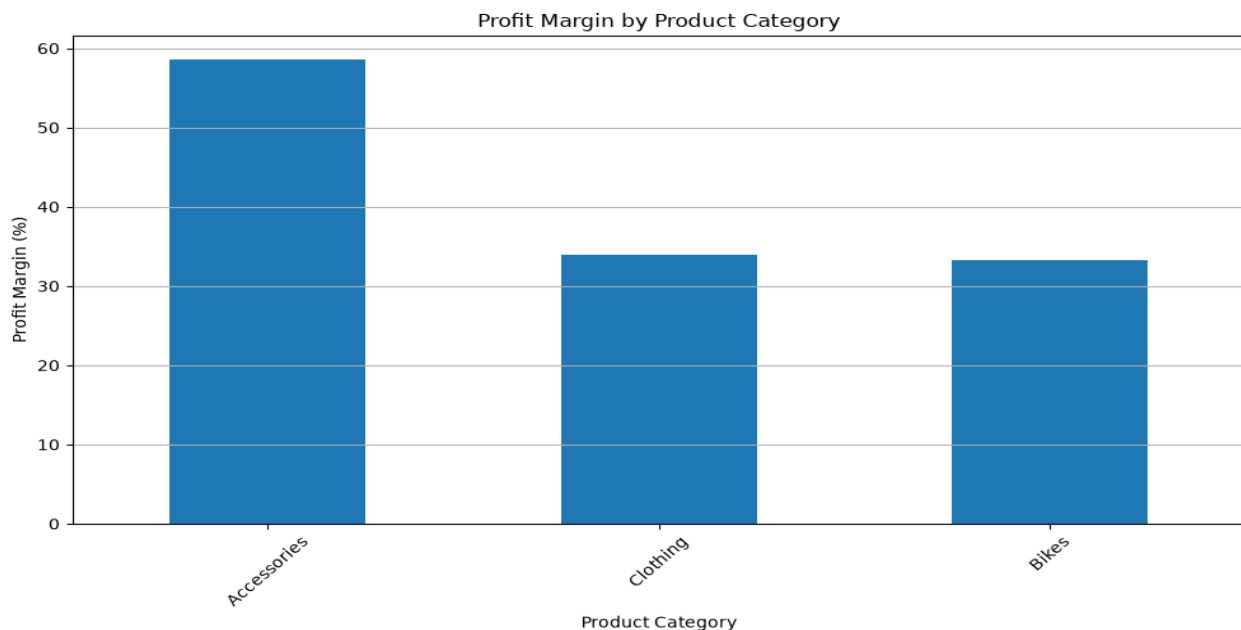


Figure 3.9: Profit Margin by Product Category

Interpretation:

Accessories have the highest profit margin at approximately 58.62%, while Clothing is approximately 33.92% and Bikes approximately 33.21%. This demonstrates that the category with the highest revenue is not necessarily the category with the highest margin.

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Results Obtained

The retail sales EDA was successfully implemented in Python using Pandas, NumPy, Matplotlib, and Seaborn. The dataset was loaded and inspected, numerical variables were summarized, time-based revenue trends were generated, customer distributions were visualized, top-selling products were identified, category revenue was compared, correlations were analyzed, and an additional profit-margin visualization was created.

The analysis produced eight documented visualizations: monthly sales trend, quarterly sales trend, age-group distribution, gender distribution, top 10 best-selling products, revenue by product category, profit margin by product category, and the correlation heatmap.

4.2 Key Findings

1. Adults (35–64) are the largest customer age group, with 55,824 records.
2. Young Adults (25–34) are the second-largest age group, with 38,654 records.
3. The gender distribution is relatively balanced: 58,312 male and 54,724 female records.
4. Water Bottle – 30 oz. is the highest-volume product with 164,086 units sold.
5. Bikes generate the highest category revenue at 61,782,134.
6. Accessories generate a much higher profit margin than Bikes and Clothing.
7. Cost and Revenue have an extremely strong positive correlation of 0.99.
8. Profit and Revenue have a strong positive correlation of 0.96.
9. Monthly and quarterly revenue show significant fluctuations and stronger peaks in later periods.

4.3 Business Recommendations

1. Prioritize high-margin Accessories through promotions, cross-selling, and product bundles.
2. Maintain adequate Bike inventory because Bikes are the largest revenue-generating category.
3. Keep high-demand products such as Water Bottle – 30 oz. and Patch Kit/8 Patches consistently stocked.
4. Use monthly and quarterly demand patterns to plan inventory, staffing, and marketing campaigns before high-demand periods.
5. Target Adults and Young Adults as primary customer segments while developing focused campaigns for Youth and Seniors.

6. Monitor cost, revenue, and profit together because the correlation analysis shows that these financial variables are closely connected.

4.4 Conclusion

The Level 1 – Task 1 Exploratory Data Analysis project successfully transformed a large retail transaction dataset into meaningful business insights. The implementation demonstrated the complete EDA workflow from data loading and inspection to statistical analysis, visualization, interpretation, and recommendations.

The findings show that Bikes are the strongest revenue category, Accessories provide the highest profit margin, and a small set of products accounts for substantial sales volume. Customer analysis identifies Adults and Young Adults as the dominant segments, while time-series analysis reveals substantial seasonal and periodic fluctuations.

Overall, the project demonstrates that Python-based EDA can support practical retail decisions related to inventory, marketing, product prioritization, and profitability. The completed notebook and visual evidence provide a strong foundation for advanced customer segmentation, predictive modeling, and dashboard development.

4.5 Future Enhancements

1. Develop an interactive Power BI or Streamlit dashboard.
2. Perform detailed customer segmentation using RFM or clustering techniques.
3. Build sales forecasting models for monthly and quarterly revenue.
4. Analyze country, state, and sub-category performance in greater detail.
5. Develop predictive models for customer purchase behavior and product demand.
6. Add automated data-quality checks and reusable EDA functions.

4.6 References

1. Project-provided retail sales CSV dataset: Customer_Segmentation_py.csv.
2. Python Pandas documentation and library usage as implemented in the project notebook.
3. Matplotlib and Seaborn visualization libraries as implemented in the project notebook.
4. Level 1 – Task 1: EDA on Retail Sales Data project requirements.